

# Purging viral latency by a bifunctional HSV-vectored therapeutic vaccine in chronically SIV-infected macaques

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## Abstract

The persistence of latent viral reservoirs remains the major obstacle to eradicating human immunodeficiency virus (HIV). We herein reported that recombinant herpes simplex virus type I (HSV-1) with ICP34.5 deletion could more effectively reactivate HIV latency than its wild-type counterpart. Mechanistically, HSV-ΔICP34.5 promoted the phosphorylation of HSF1 by decreasing the recruitment of protein phosphatase 1 (PP1α), thus effectively binding to the HIV LTR to reactivate the latent reservoirs. In addition, HSV-ΔICP34.5 enhanced the phosphorylation of IKKα/β through the degradation of IκBα, leading to p65 accumulation in the nucleus to elicit NF-κB pathway-dependent reactivation of HIV latency. Then, we constructed the recombinant HSV-ΔICP34.5 expressing simian immunodeficiency virus (SIV) env, gag, or the fusion antigen sPD1-SIVgag as a therapeutic vaccine, aiming to achieve a functional cure by simultaneously reactivating viral latency and eliciting antigen-specific immune responses. Results showed that these constructs effectively elicited SIV-specific immune responses, reactivated SIV latency, and delayed viral rebound after the interruption of antiretroviral therapy (ART) in chronically SIV-infected rhesus macaques. Collectively, these findings provide insights into the rational design of HSV-vectored therapeutic strategies for pursuing an HIV functional cure.

### eLife Assessment

In this **useful** study, the authors tested a novel approach to eradicating HIV reservoirs by constructing a herpes simplex virus (HSV)-based therapeutic vaccine and evaluating efficacy in experimental infections of chronically SIV-infected, antiretroviral therapy (ART)-treated macaques. While mean viremia at rebound was lower in the HSV vaccine-treated group, the evidence presented appears to be **incomplete**, as the group size was small and the viral load at rebound was highly variable. This is a revised paper, but the support for the conclusions, particularly the effect of the HSV-vectored therapeutic vaccine on the SIV reservoir in the SIV-infected macaques, remains limited.

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## Introduction

The epidemic of acquired immunodeficiency syndrome (AIDS), caused by human immunodeficiency virus type I (HIV-1), is still a huge challenge for global public health, with approximately 39 million people living with HIV-1 as of 2022. To date, there is neither a curable drug nor a prophylactic vaccine for clinical use against HIV-1 infections (Sun et al., 2023b [↗](#), Sun et al., 2023a [↗](#)). Antiretroviral therapy (ART) can effectively control HIV-1 replication to an undetectable level, but the termination of ART usually results in prompt viral rebound from latent viral reservoirs (Archin et al., 2014 [↗](#), Looker et al., 2017 [↗](#), Calistri et al., 2003 [↗](#), Heng et al., 1994 [↗](#)). Thus, it is of great priority to explore novel strategies for curing HIV latency. The “Shock and Kill” strategy is considered a promising approach for purging HIV-1 reservoirs, involving the activation of latently infected cells to express viral products (Shock), followed by viral cytopathic effects or specific cytolytic T lymphocytes (CTLs) to eliminate the activated cells (Kill) (Wu et al., 2022 [↗](#), Yang et al., 2019 [↗](#), Kim et al., 2018 [↗](#)). Numerous latency-reversing agents (LRAs) (Yang et al., 2019 [↗](#), Wu et al., 2022 [↗](#)), including methylation inhibitors, histone deacetylase (HDAC) inhibitors (Archin et al., 2017 [↗](#), Lehrman et al., 2005 [↗](#)), and bromodomain and extra terminal domain (BET) protein inhibitors (Li et al., 2013 [↗](#), Bisgrove et al., 2007 [↗](#)), have been identified to reactivate latent HIV-1 in preclinical studies, but there is no ideal LRA available for clinical patients yet.

Herpes simplex virus (HSV), a human herpesvirus, features a 152-kb double-stranded DNA genome encoding over 80 proteins (Poh, 2016 [↗](#)). Owing to its distinctive genetic background, high capacity, broad tropism, thermostability, and excellent safety profile, the modified HSV constructs have extensive applications in gene therapy and oncolytic virotherapy. For example, talimogene laherparepvec (T-VEC), an HSV-1 variant with ICP34.5 deletion and GM-CSF insertion, received FDA approval in 2015 for treating malignant melanoma, showcasing notable safety and efficacy in clinical practice. Recombinant HSV-based constructs have also emerged as efficacious gene delivery vectors against infectious diseases. Early studies indicated that prophylactic vaccines based on HSV, expressing simian immunodeficiency virus (SIV) antigens, could elicit robust antigen-specific CTL responses in mice and monkeys, providing enduring and partial protection against pathogenic SIVmac239 challenges (Kaur et al., 2007 [↗](#), Murphy et al., 2000 [↗](#)). Moreover, increasing data suggest the crucial role of HIV-specific CTL in controlling viral replication and eliminating potential HIV reservoirs (Collins et al., 2020 [↗](#), Leitman et al., 2017 [↗](#)). Significantly, epidemiological research suggests a synergistic effect between HSV and HIV infections, with HSV infection in HIV patients being associated with increased HIV-1 viral load and disease progression (Looker et al., 2017 [↗](#), Calistri et al., 2003 [↗](#), Heng et al., 1994 [↗](#)). Some studies have further

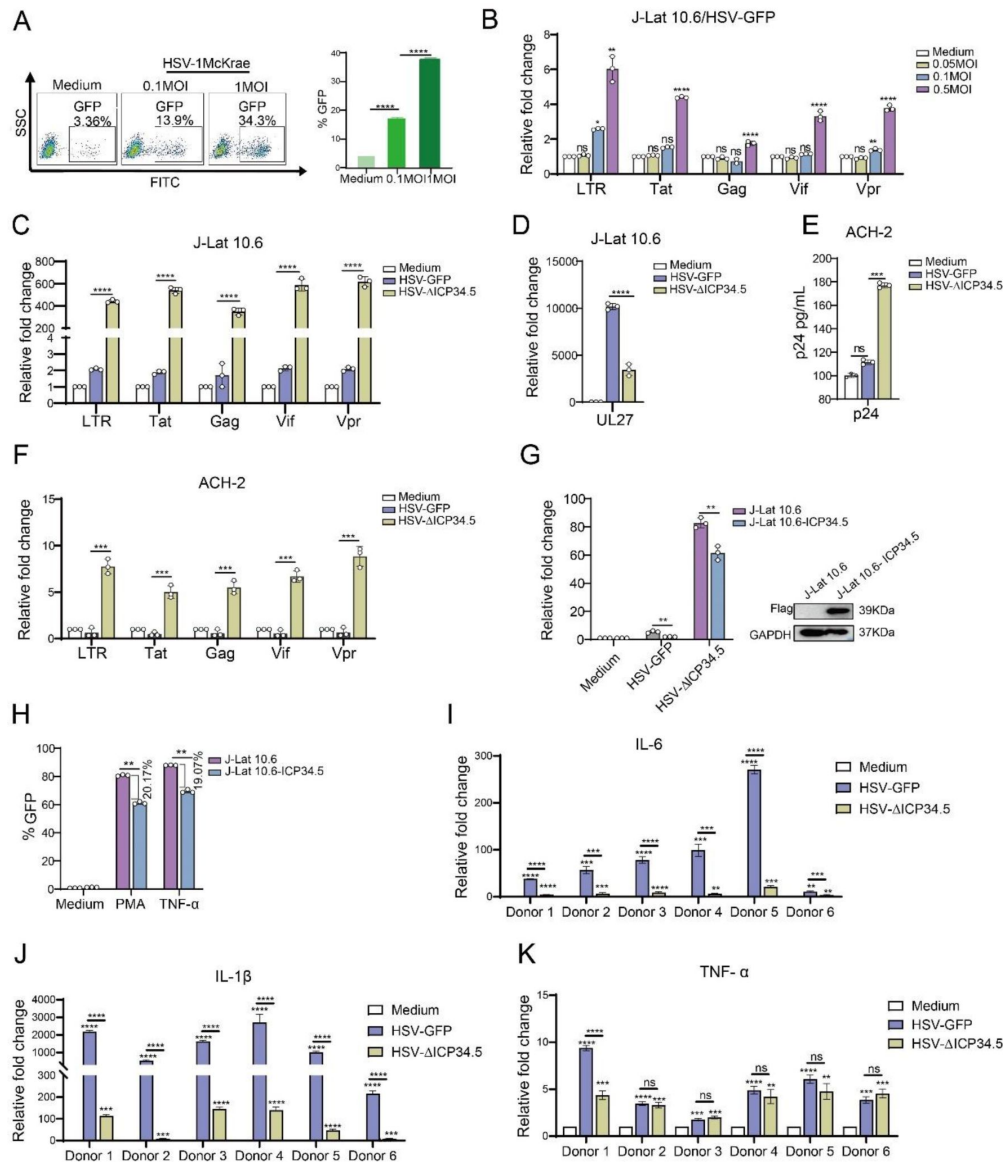
unveiled the ability of HSV to activate HIV latent reservoirs (Amici et al., 2004 [↗](#), Amici et al., 2001 [↗](#), Pierce et al., 2023 [↗](#)). Given the potential of HSV to simultaneously induce antigen-specific immune responses and reactivate latent viral reservoirs, we propose a proof-of-concept strategy to achieve an HIV functional cure using a modified bifunctional HSV-vectored therapeutic vaccine.

## Results

### The modified HSV-ΔICP34.5-based constructs reactivated HIV latency more efficiently than wild-type HSV counterparts

The J-Lat 10.6 cell line, derived from Jurkat T cells containing latent HIV-1 provirus, was infected with wild-type HSV-1 Mckrae strain at different multiplicities of infection (MOI) to assess its capability to reactivate HIV latency. Flow cytometry analysis showed a dose-dependent increase in green fluorescent protein (GFP) expression (**Figure 1A** [↗](#)), indicating that HSV-1 can reactivate latent HIV. Subsequently, we infected J-Lat 10.6 cells with an HSV-1 17 strain containing GFP (HSV-GFP) and found significant upregulation of mRNA levels of HIV-1 LTR, Tat, Gag, Vif, and Vpr, demonstrating that the attenuated HSV-1 strain can reactivate latent HIV at different MOIs (**Figure 1B** [↗](#)). Using the bacterial artificial chromosome (BAC)/galactokinase (galK) system, we then constructed the recombinant HSV deleted with ICP34.5 (HSV-ΔICP34.5) based on HSV-GFP. Notably, we discovered for the first time that the HSV-ΔICP34.5 reactivated HIV latency more efficiently than its parental strain (HSV-GFP). Specifically, mRNA levels of HIV genes (LTR, Tat, Gag, Vpr, Vif) were significantly higher in J-Lat 10.6 cells infected with HSV-ΔICP34.5 compared to those treated with HSV-GFP (**Figure 1C** [↗](#)), despite the weaker replication ability of HSV-ΔICP34.5 in these cells, as indicated by the mRNA level of HSV UL27 (**Figure 1D** [↗](#), **Figure S1**).

Furthermore, this finding was validated in ACH-2 cells, which are derived from T cells latently infected with replication-competent HIV-1. A significantly higher level of p24 protein, a key indicator of HIV replication, was found in the HSV-ΔICP34.5-infected ACH-2 cells compared to HSV-GFP-treated cells (**Figure 1E** [↗](#)), and the mRNA levels of HIV-related genes (LTR, Tat, Gag, Vpr, Vif) were also significantly increased (**Figure 1F** [↗](#)). Subsequently, we generated J-Lat 10.6 cells stably expressing ICP34.5-Flag-Tag (J-Lat 10.6-ICP34.5) using a recombinant lentivirus system and confirmed the expression of the ICP34.5 protein. Our research revealed that HSV-ΔICP34.5 displayed a reduced capacity to reverse HIV latency in J-Lat 10.6-ICP34.5 cells compared to J-Lat 10.6 cells (**Figure 1G** [↗](#)). Moreover, in J-Lat 10.6-ICP34.5 cells, the potency of latent reversal agents such as phorbol 12-myristate 13-acetate (PMA) and TNF-α was notably reduced when contrasted with J-Lat 10.6 cells (**Figure 1H** [↗](#)). We also confirmed the enhanced reactivation of HIV latency by HSV-ΔICP34.5 in primary CD4<sup>+</sup> T cells from people living with HIV (PLWH) (**Figure S2**). In addition, HSV-ΔICP34.5 induced a lower level of inflammatory cytokines (including IL-6, IL-1β, and TNF-α) in primary CD4<sup>+</sup> T cells from PLWH compared to HSV-GFP stimulation, likely due to its lower virulence and replication ability (**Figure 1I-K** [↗](#)). Thus, these data suggested that the safety of HSV-ΔICP34.5 in PLWH might be tolerable. In addition, both adenovirus and vaccinia virus cannot reactivate HIV latency in this study (**Figure S3**), and the deletion of ICP0 gene from HSV-1 diminished the reactivation effect of HIV latency by HSV-1 (**Figure S4**), implying this reactivation might be virus-specific. Overall, these findings indicate that the recombinant HSV-ΔICP34.5 construct specifically reactivate HIV latency with high efficiency.



**Figure 1.**

### The modified HSV-ΔICP34.5-based constructs reactivated HIV latency more efficiently than wild-type HSV counterparts.

(A) J-Lat 10.6 cells ( $1 \times 10^6$ ) were infected with varying MOIs of wild-type HSV-1 Mckrae strain for 30 h. The proportion of GFP+ cells, indicating activated latent cells, is shown in the pseudocolor plot (left) and the corresponding bar chart (right). (B) J-Lat 10.6 cells ( $1 \times 10^6$ ) were infected with varying MOIs of HSV-1 17 strain containing GFP (HSV-GFP) for 30 h, and then the mRNA levels of HIV-1 LTR, Tat, Gag, Vpr, Vif are shown with the histogram. (C) J-Lat 10.6 cells ( $1 \times 10^6$ ) were infected with HSV-GFP or HSV-ΔICP34.5 at an MOI of 0.1 for 30 h. The mRNA levels of HIV-1 LTR, Tat, Gag, Vpr, Vif, and (D) HSV-1 UL27 are shown with the histogram. (E) ACH-2 cells were infected with HSV-GFP or HSV-ΔICP34.5 at an MOI of 0.1 for 30 h. The p24 protein level was detected using an HIV-1 p24 ELISA kit, and the mRNA levels of HIV-1 LTR, Tat, Gag, Vpr, and Vif are shown in the histogram (F). (G) J-Lat 10.6 and J-Lat 10.6-ICP34.5 cells were infected with HSV-GFP or HSV-ΔICP34.5, and the mRNA levels of HIV-1 Tat were shown with the histogram (left). Blotting showed that J-Lat 10.6 cells stably expressing HSV ICP34.5 (J-Lat 10.6-ICP34.5) can appropriately express ICP34.5 protein using Flag-tag antibodies (right). (H) J-Lat 10.6 and J-Lat 10.6-ICP34.5 cells were respectively stimulated with PMA (10 ng/mL) and TNF- $\alpha$  (10 ng/mL), and the expression level of GFP+ cells is displayed with the corresponding bar chart. (I-K) Primary CD4+ T cells from people living with HIV (PLWH) were infected with HSV-GFP or HSV-ΔICP34.5. The inflammatory response was assessed by evaluating mRNA levels of IL-6, IL-1 $\beta$  and TNF- $\alpha$  using qPCR. Data shown are mean  $\pm$  SD. \*\* $P < 0.01$ , \*\*\* $P < 0.001$ , \*\*\*\* $P < 0.0001$ . ns: no significance.

## The modified HSV-based constructs effectively reactivated HIV latency by modulating the IKK $\alpha$ / $\beta$ -NF- $\kappa$ B pathway and PP1-HSF1 pathway

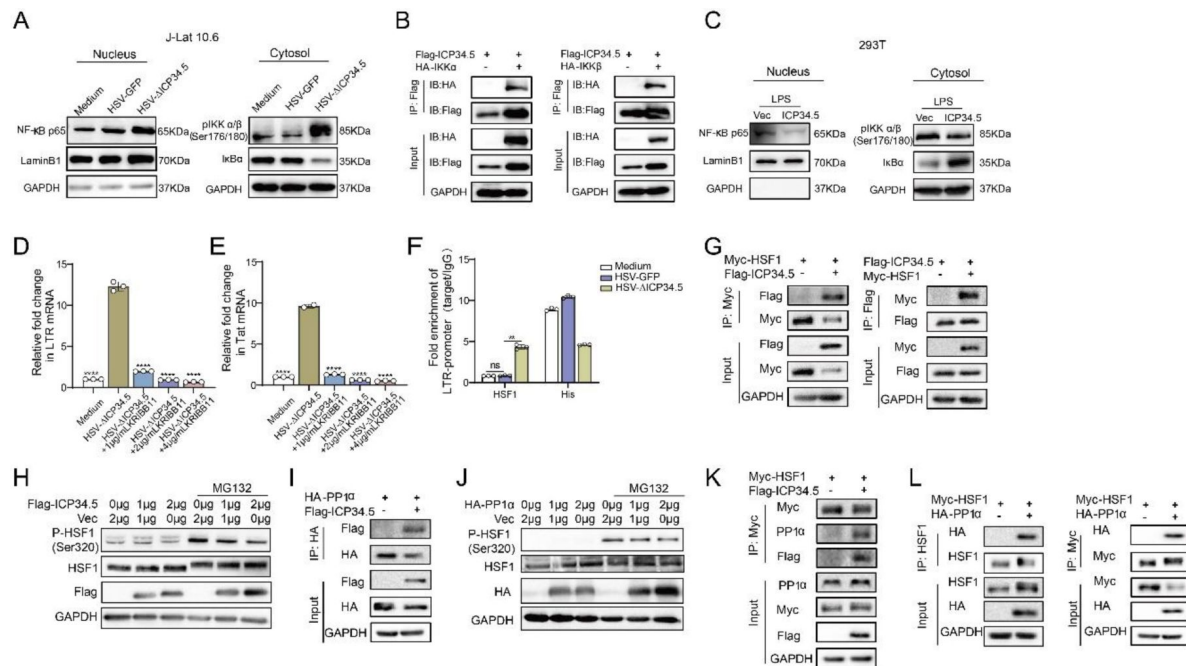
Next, based on RNA-seq analysis for HSV- $\Delta$ ICP34.5-induced signaling pathways (Figure S5) and previous literature, we explored the mechanism of reactivating viral latency by HSV- $\Delta$ ICP34.5-based constructs. J-Lat10.6 cells were infected with HSV-GFP or HSV- $\Delta$ ICP34.5, and results showed that HSV- $\Delta$ ICP34.5 significantly enhanced the phosphorylation of IKK $\alpha$ / $\beta$ , promoted the degradation of I $\kappa$ B $\alpha$ , and thus led to the accumulation of p65 in the nucleus (Figure 2A). NF- $\kappa$ B is a well-known host transcription factor that exists in the form of the NF- $\kappa$ B-I $\kappa$ B complex in resting cells, but I $\kappa$ B can degrade and release the NF- $\kappa$ B dimer to enter the nucleus and promote gene transcription in response to external stimulation. Using the coimmunoprecipitation (Co-IP) assay, we verified that the ICP34.5 protein had a specific interaction with IKK $\alpha$ / $\beta$ , and then ICP34.5 could dephosphorylate IKK $\alpha$ / $\beta$ . Moreover, the overexpression of ICP34.5 effectively inhibited lipopolysaccharide (LPS)-induced NF- $\kappa$ B pathway activation by inhibiting p65 entry into the nucleus (Figure 2B-C).

To further clarify the underlying mechanism, we performed the immunoprecipitation-mass spectrometry (IP-MS) analysis in ICP34.5 overexpressing cells to identify other potential molecules contributing to this reactivation, and we found that ICP34.5 can also interact with heat shock 1 protein (HSF1) (Table S1). HSF1 has been reported as a transcription factor correlated with the reactivation of HIV latency (Xu et al., 2022, Lin et al., 2018, Zeng et al., 2017). To test whether HSF1 contributes to the reactivation of HIV latency by HSV- $\Delta$ ICP34.5-based constructs, KRIBB11, an inhibitor of HSF1, was administered to HSV- $\Delta$ ICP34.5-infected J-Lat 10.6 cells. The results indicated that the reactivation ability of HSV- $\Delta$ ICP34.5 was significantly inhibited by KRIBB11 treatment in a dose-dependent manner (Figure 2D-E). Furthermore, a significant enhancement of the binding of HSF1 to the HIV LTR was observed upon HSV- $\Delta$ ICP34.5 infection, leading to an increase in the reactivation of HIV latency (Figure 2F). The direct interaction between ICP34.5 and HSF1 was also identified by Co-IP assay. Importantly, HSF1 was effectively dephosphorylated at Ser320 as a result of the overexpression of ICP34.5, while no influence on the level of HSF1 expression was observed (Figure 2G-H, Figure S6). Considering that protein phosphatase 1 (PP1) can interact with ICP34.5 and dephosphorylate eIF2 $\alpha$  (Li et al., 2011), we then investigated the interaction between PP1 $\alpha$  and ICP34.5 (Figure 2I). Additionally, a direct interaction between PP1 $\alpha$  and HSF1 was found, allowing for the dephosphorylation of HSF1 and then affecting its ability to reactivate HIV latency (Figure 2J-L). Collectively, these findings demonstrated that our modified HSV- $\Delta$ ICP34.5-based constructs effectively reactivated HIV latency by modulating the IKK $\alpha$ / $\beta$ -NF- $\kappa$ B pathway and PP1-HSF1 pathway.

## Recombinant HSV- $\Delta$ ICP34.5 constructs expressing SIV antigens elicited robust immune responses in mice

Subsequently, we explored the potential of HSV- $\Delta$ ICP34.5 as a bifunctional therapeutic vector to not only reactivate latent viral reservoirs but also induce antigen-specific immune responses against viral replication. To achieve this, SIV antigen genes were introduced into HSV- $\Delta$ ICP34.5 vector based onBAC/ galK system. Additionally, the ICP47 gene was ablated to augment the immunogenicity of the HSV vector (Figure 3A, Figure S7). A series of recombinant HSV- $\Delta$ ICP34.5 $\Delta$ ICP47-based vectors expressing SIV gag and env antigen were constructed, and the antigen expression of these constructs was confirmed by Western blotting assay (Figure 3B and C). Furthermore, to improve the immunogenicity of the targeted antigen, we fused the SIV gag with soluble PD1 (sPD1), enabling it to competitively bind with PD-L1 and thereby block the PD1/PD-L1 immune inhibitory pathway (Figure 3D). Consistent with the above findings, these HSV- $\Delta$ ICP34.5 $\Delta$ ICP47-based SIV vaccines also efficiently reactivated HIV latency (Figure S8). Then, the immunogenicity of the above modified HSV- $\Delta$ ICP34.5 $\Delta$ ICP47-based SIV vaccine was assessed in





**Figure 2.**

### The modified HSV-based constructs effectively reactivated HIV latency by modulating the NF-κB pathway and HSF1 pathway.

(A) J-Lat 10.6 cells infected with HSV-GFP and HSV-ΔICP34.5 at MOI 0.1. Cytoplasmic and nuclear proteins were analyzed for p65, p-IKKα/β and IκBα levels. GAPDH and Lamin B1 served as loading controls for cytoplasmic and nuclear proteins, respectively. (B) 293T cells transfected with Flag-ICP34.5, IKKα (left), or IKKβ (right) analyzed through Co-IP assays. (C) 293T cells were transfected with Flag-ICP34.5 or empty vector (Vec) for 24h, treated with LPS (1μg/mL) for 8h. Cytoplasmic and nuclear proteins analyzed by Western blot (WB). (D-E) J-Lat 10.6 cells infected with HSV-ΔICP34.5 treated with various KRIBB11 concentrations. LTR and Tat mRNA levels analyzed by qPCR. (F) J-Lat 10.6 cells infected with HSV-wt or HSV-ΔICP34.5 at MOI 0.1 for 36h. ChIP-qPCR assessed HSF1 binding to LTR. IgG and Histone antibody (His) served as negative and positive controls. (G) 293T cellstransfected with Flag-ICP34.5 and Myc-HSF1, analyzed by Co-IP assays. (H) The immunoblot depicts the alterations in protein levels in 293T cells transfected with either empty vector or ICP34.5 for 6 h, followed by 24 h of treatment with 10 μM MG132. (I) 293T cells transfected with Flag-ICP34.5 and HA-PP1α, analyzed by Co-IP assays. (J) The immunoblot depicts the alterations in protein levels in 293T cells transfected with either empty vector or HA-PP1α for 6 h, followed by 24 h of treatment with 10 μM MG132. (K) 293T cells transfected with Myc-HSF1 with or without Flag-ICP34.5, analyzed by Co-IP assays. (L) 293T cells transfected with Myc-HSF1 and HA-PP1α, analyzed through Co-IP assays. Data shown are mean ± SD. \*\* $P < 0.01$ , \*\*\*\* $P < 0.0001$ . ns: no significance.

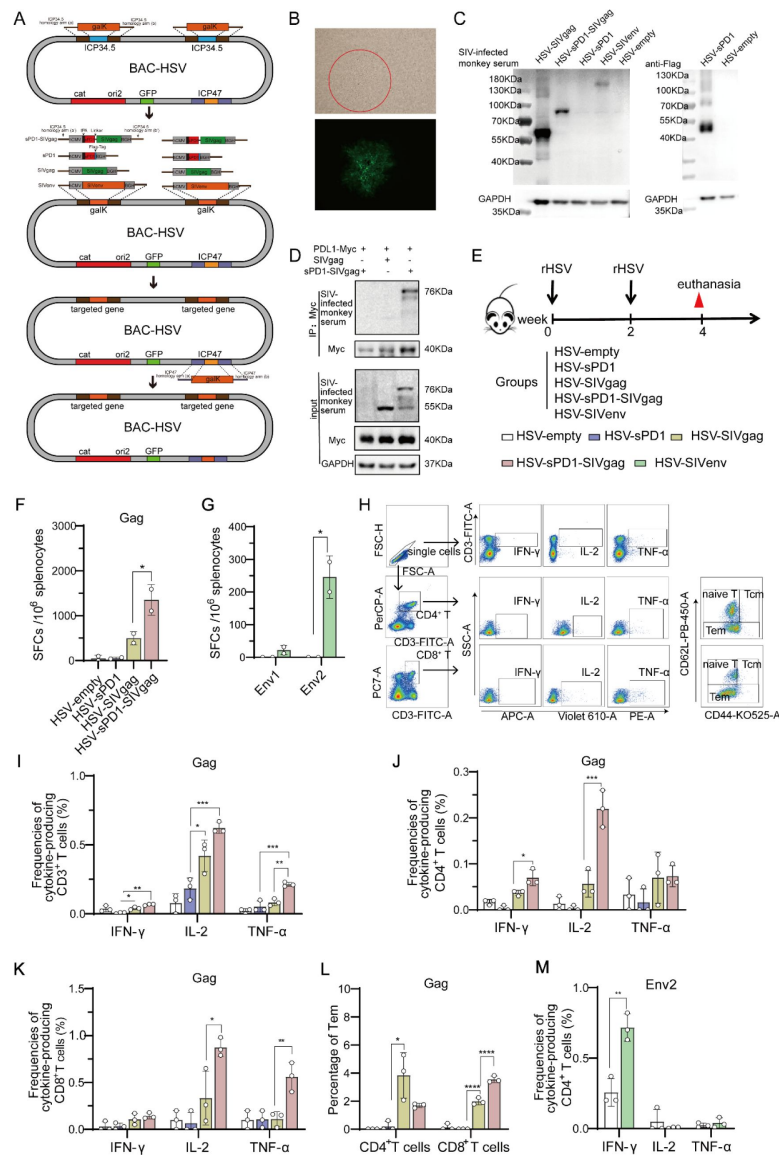
mice (**Figure 3E**). Our results showed that these constructs effectively elicited SIV antigen-specific T cell immune responses using the interferon  $\gamma$  (IFN- $\gamma$ ) ELISpot assay and the intracellular cytokine staining (ICS) assay. Of note, the frequency of SIV Gag-specific IFN- $\gamma$ -secreting spot-forming cells (SFCs) in the HSV-sPD1-SIVgag group (1350 SFCs per  $10^6$  splenocytes) was significantly higher than that in the HSV-SIVgag group (498 SFCs per  $10^6$  splenocytes) (**Figure 3F**). The frequency of SIV Env2-specific IFN- $\gamma$ -secreting SFCs in the HSV-SIVenv group was significantly higher than the HSV-empty group (**Figure 3G**). Furthermore, the polyfunctionality of antigen-specific T cell subsets in response to SIV antigen stimulation was confirmed using the ICS assay (**Figure 3H-M**). Consistently, the HSV-sPD1-SIVgag group showed a significantly higher frequency of SIV-specific CD3<sup>+</sup> T, CD4<sup>+</sup> T, and CD8<sup>+</sup> T cell subsets secreting IFN- $\gamma$ , IL-2, and TNF- $\alpha$  cytokines compared to the HSV-SIVgag group (**Figure 3I-K**). Notably, a heightened proportion of Gag-specific effector memory T cells (Tem) of CD8<sup>+</sup> T cell subset in HSV-sPD1-SIVgag group was observed in comparison to the HSV-Gag group (**Figure 3L**). In addition, a higher frequency of SIV-Env2-specific CD4<sup>+</sup> T cells secreting IFN- $\gamma$  was observed in the HSV-SIVenv group than in the HSV-empty group (**Figure 3M**). These data indicated that the vaccines constructed in this study elicited a robust T cell immune response in mice. Moreover, the blockade of PD1/PDL1 signaling pathway effectively enhanced vaccine-induced T cell immune responses, which was consistent with our and other previous studies (Zhou et al., 2013, Wu et al., 2022, Pan et al., 2018, Xiao et al., 2014).

### The modified HSV-based constructs efficiently elicited SIV-specific immune responses in chronically SIV-infected macaques

Next, the immunogenicity of these HSV-vectored SIV vaccines was further investigated in chronically SIV<sub>mac239</sub>-infected rhesus macaques (RMs). To mimic chronically infected HIV patients in clinic practice, all RMs used in this study were chronically infected with SIV and received ART treatment for several years, as reported in our previous studies (Pan et al., 2018, Yang et al., 2019, Wu et al., 2021, Wu et al., 2022, He et al., 2023). Based on sex, age, viral load, and CD4 count, nine RMs were assigned into three groups: ART+saline group (n=3), ART+HSV-empty group (n=3), and ART+HSV-sPD1-SIVgag/SIVenv group (n=3) (Table S2). All RMs received ART (FTC/6.7 mg/kg/once daily, PMPA/10 mg/kg/once daily) treatment to avoid the interference of free SIV particles. On day 33 and day 52, these RMs were immunized with saline, HSV-empty, or HSV-sPD1-SIVgag/SIVenv respectively. On day 70, ART treatment in all RMs was discontinued to evaluate the time interval of viral rebound. Samples were collected at different time points to monitor virological and immunological parameters (**Figure 4A**). To reduce the impact of individual RM variations, the difference in SIV-specific IFN- $\gamma$ -secreting SFCs between post-immunization and pre-immunization ( $\Delta$ SFCs) was used to evaluate the immune response induced by HSV-vectored SIV vaccines. The results showed that SIV Gag-specific  $\Delta$ SFCs in the ART+HSV-sPD1-SIVgag/SIVenv group were greatly increased when compared with those in the ART+HSV-empty group and ART+saline group (**Figure 4B-D**). A similar enhancement of SIV Gag-specific TNF- $\alpha$ -secreting CD4<sup>+</sup> T and CD8<sup>+</sup> T subsets was also verified by ICS assay (**Figure 4E**). Collectively, these data demonstrated that the HSV-sPD1-SIVgag/SIVenv construct elicited robust SIV-specific T cell immune responses in ART-treated, SIV-infected RMs.

### The modified HSV-based constructs effectively reactivated SIV latency *in vivo* and delayed viral rebound in chronically SIV-infected, ART-treated macaques

Finally, we investigated the therapeutic efficacy of HSV-sPD1-SIVgag/SIVenv in chronically SIV-infected, ART-treated RMs. Consistent with our previous studies, the plasma viral load (VL) in these RMs was effectively suppressed during ART treatment, but rebounded after ART discontinuation. The VL in the ART+saline group promptly rebounded after ART discontinuation, with an average

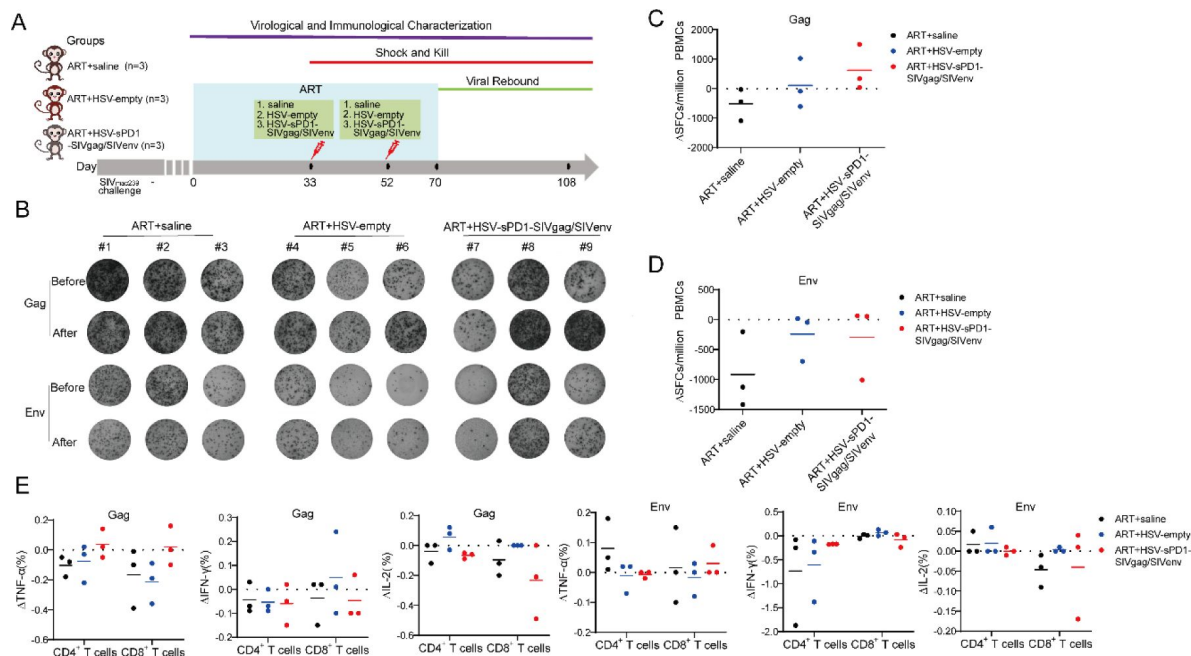


**Figure 3.**

### Recombinant HSV-1 vector-based SIV vaccines induce specific T cell immune responses in mice.

(A) Schematic diagram illustrating the construction of recombinant HSV through the BAC/gaK selection system. The ICP34.5 gene was replaced with the gaK gene via homologous recombination, followed by substituting gaK with a target gene expression cassette containing the hCMV promoter and BGH terminator. Finally, the ICP47 gene was deleted. (B) Brightfield (top) and fluorescence (bottom) images of a clone of the rescued recombinant HSV. (C) Vero cells infected with recombinant HSV constructs, with protein expression of targeted genes detected using SIV-infected monkey serum. (D) HeLa cells transfected with Myc-PDL1 and then infected with HSV-empty, HSV-SIVgag, or HSV-sPD1-SIVgag at an MOI of 0.1 for 24 h. Cell lysates were subjected to Co-IP analysis. (E) Schematic schedule of mouse vaccination. Twenty-five mice were randomly allocated to five groups: HSV-empty, HSV-sPD1, HSV-SIVgag, HSV-sPD1-SIVgag, and HSV-SIVenv. Mice were injected with the corresponding vaccines at weeks 0 and 2. At week 4, mice were sacrificed, and spleen lymphocytes were collected to evaluate immune response. (F-G) Column graphs showing the number of Gag or Env1, Env2-specific spot-forming cells (SFCs) per  $10^6$  spleen lymphocytes, as measured by IFN- $\gamma$  ELISpot assay. (H) Pseudocolor plot of flow cytometry illustrating the gating strategy. Column graphs showing the frequencies of IFN- $\gamma$ , IL-2, and TNF- $\alpha$  production from gag-specific CD3 $^+$  T (I), CD4 $^+$  T (J), and CD8 $^+$  T cells (K). (L) Bar chart showing the proportion of Tem (effector memory T cells) among CD4 $^+$  T and CD8 $^+$  T cells upon stimulation with the SIV Gag peptide pools. (M) Bar chart showing the frequencies of Env2-specific IFN- $\gamma$  $^+$  CD4 $^+$  T cells. Data were expressed as mean $\pm$ SD from five mice samples. Three independent experiments for the animal immunization were repeated. \* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ , \*\*\*\* $P < 0.0001$ . ns: no significance.





**Figure 4.**

### The modified HSV-based constructs efficiently elicited SIV-specific immune responses in chronically SIV-infected macaques.

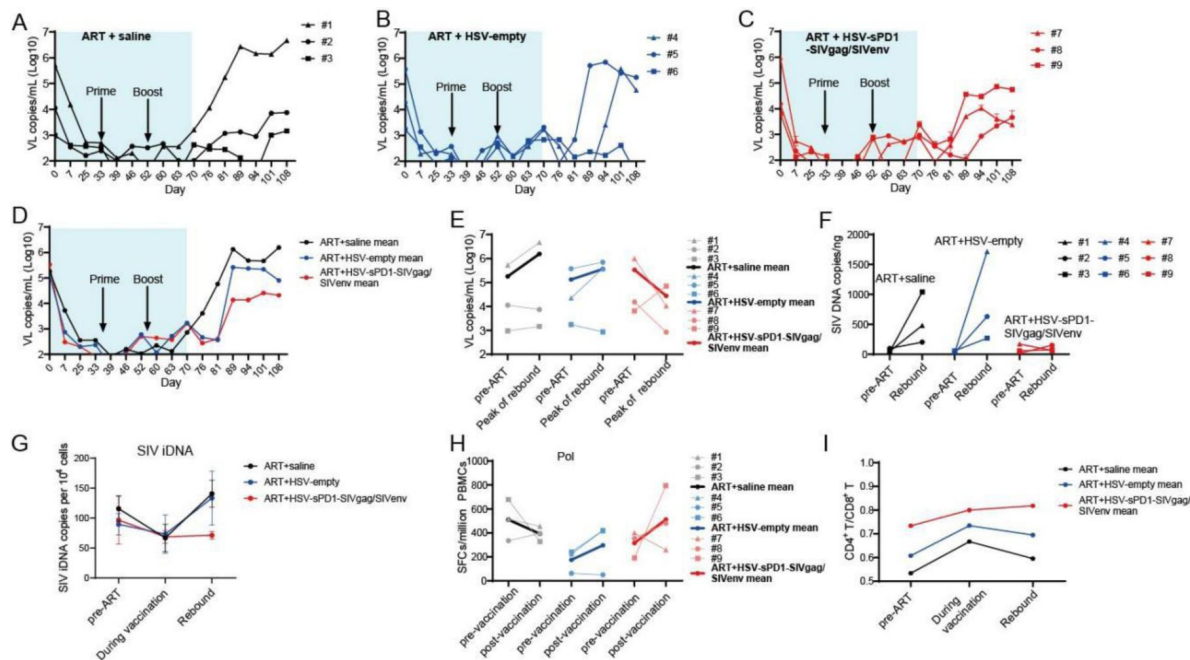
**(A)** Schematic schedule of the macaque experiment. Nine chronically SIV-infected macaques were assigned into three groups: ART+saline group (n=3), ART+HSV-empty group (n=3), and ART+HSV-sPD1-SIVgag/SIVenv group (n=3). All SIV-infected macaques received ART treatment (FTC/6.7 mg/kg/once daily, PMPA/10 mg/kg/once daily) for 33 days. On days 33 and 52, macaques were immunized with saline, HSV-empty, and HSV-sPD1-SIVgag/SIVenv respectively. ART treatment, was interrupted in all macaques on day 70 after the second vaccination. Samples were collected at various time points to monitor virological and immunological parameters. **(B)** Representative images of Gag or Env-specific spots (2.5×10<sup>5</sup> cells per well) from each macaque pre-vaccination (before, day 33) and post-vaccination (after, day 70) by ELISpot assay. **(C-D)** Difference in SIV-specific IFN-γ-secreting cells (ΔSFCs) between pre- and post-immunization, assessing the immune response induced by HSV-vectored SIV vaccines. **(E)** Difference in SIV-specific TNF-α/IFN-γ/IL-2-secreting CD4<sup>+</sup> T and CD8<sup>+</sup> T subsets between pre- and post-immunization, detected by ICS assay.

8.63-fold increase in the rebounded peak VL compared with the pre-ART VL (**Figure 5A** [↗](#), *D and E*). However, plasma VL in the ART+HSV-empty group and the ART+HSV-sPD1-SIVgag/SIVenv group exhibited a delayed rebound interval (**Figure 5B-D** [↗](#)). Remarkably, there was a lower rebounded peak VL than pre-ART VL in the ART+HSV-sPD1-SIVgag/SIVenv group (average 12.20-fold decrease), while a higher rebounded peak VL than pre-ART VL in the ART+HSV-empty group (average 2.74-fold increase) (**Figure 5E** [↗](#)). Then, we assessed the potential effect on the latent SIV reservoirs *in vivo* by administering our modified HSV-based SIV therapeutic constructs in these RMs. Although there were no obvious viral blips observed in these RMs, we found significant suppression of total SIV DNA and integrated SIV DNA provirus in the ART+HSV-sPD1-SIVgag/SIVenv group. However, the copies of the SIV DNA provirus were significantly improved in the ART+HSV-empty group and ART+saline group (**Figure 5F-G** [↗](#)). More interestingly, we assessed the magnitude of SIV Pol antigen-specific immune responses, which could represent to some extent the level of newly generated virions from the reactivated SIV reservoirs, because SIV Pol antigen was not included in our designed vaccine constructs. Specifically, the Pol-specific SFCs amount on Day 70 (511 SFCs/10<sup>6</sup> PBMCs, post-vaccination) was higher than that on Day 33 (315 SFCs/10<sup>6</sup> PBMCs, pre-vaccination) in the ART+HSV-sPD1-SIVgag/SIVenv group. In addition, there was a similar observation in the ART+HSV-empty group. In contrast, the Pol-specific SFCs gradually decreased with the duration of ART treatment in the ART+saline group (**Figure 5H** [↗](#)). In addition, the CD4<sup>+</sup>/CD8<sup>+</sup> T cell ratio (**Figure 5I** [↗](#)) and body weight (*Figure S9*) after treatment were effectively ameliorated in the RMs of the ART+HSV-sPD1-SIVgag/SIVenv group, but not in the ART+HSV-empty group and ART+saline group. Our data also demonstrated that there was no significant effect on the cell composition of peripheral blood in the macaques of ART+HSV-sPD1-SIVgag/SIVenv group (*Figure S10*). Taken together, these findings suggested that the latent SIV reservoirs might be effectively purged because of the effect of simultaneously reactivating viral latency and eliciting SIV-specific immune responses by our modified HSV-based SIV therapeutic constructs, thus resulting in a delayed viral rebound in chronically SIV-infected, ART-treated macaques.

## Discussion

To conquer the continuous epidemic of AIDS, exploring novel strategies to render and eliminate HIV latency should stand as a pivotal pursuit. Currently, numerous strategies, including shock and kill, block and lock, chimeric antigen receptor T-cell therapy, therapeutic vaccination, and gene editing, have been extensively investigated to target the latent HIV reservoirs for an HIV functional cure (Deeks, 2012 [↗](#), Yeh and Ho, 2021 [↗](#), Maldini et al., 2020 [↗](#), Herzig et al., 2019 [↗](#), Dash et al., 2023 [↗](#), Dashti et al., 2023 [↗](#), Walker-Sperling et al., 2022 [↗](#)). However, there is no safe and effective approach for clinical use in HIV patients yet. In the present study, we occasionally found that the modified HSV-ΔICP34.5-based constructs could reactivate HIV latency more efficiently than wild-type HSV counterpart, which inspired us to develop a proof-of-concept strategy based on a bifunctional HSV-vectored therapeutic vaccine, aiming to simultaneously reactivate viral latency and elicit HIV/SIV-specific immune responses for HIV functional cure. Our results indicated that these modified HSV-based constructs efficiently elicited antigen-specific immune responses in mice and chronically SIV-infected macaques, and further therapeutic efficacy experiments showed that this strategy effectively reactivated SIV latency *in vivo* and delayed viral rebound in chronically SIV-infected, ART-treated macaques (**Figure 6** [↗](#)).

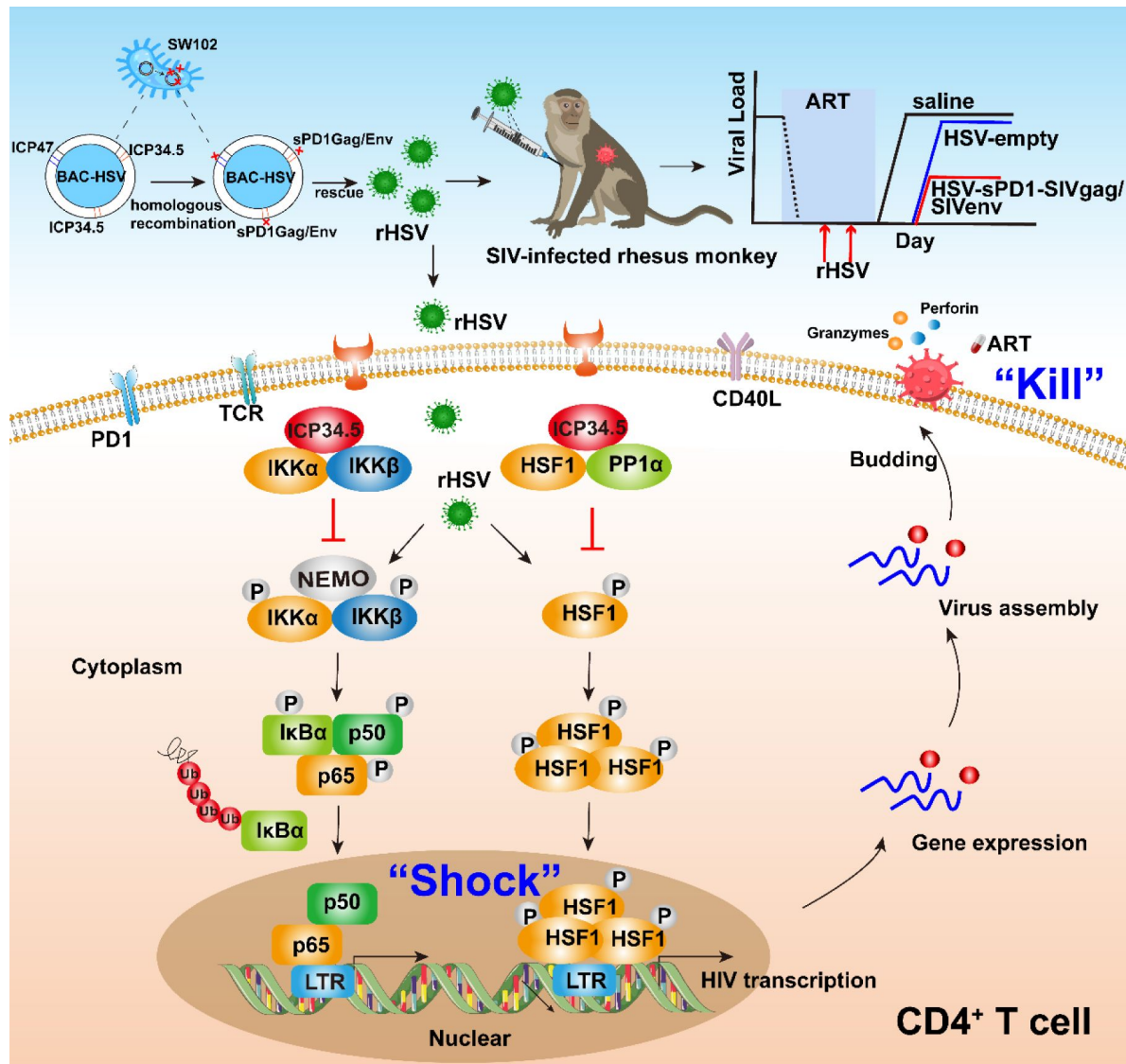
The latent HIV proviruses can harbor it into the host genome with a quiescent transcription state, and thus cannot be recognized by immune surveillance or drug killing (Churchill et al., 2016 [↗](#), Pierson et al., 2000 [↗](#)). Therefore, it is critical to disrupt viral latency for developing an HIV cure strategy. Based on our experimental data, the mechanism for efficiently reactivating viral latency by the modified HSV-ΔICP34.5-based constructs may involve regulating the IKKα/β-NF-κB pathway and PP1-HSF1 pathway. Indeed, during its replication, HSV can activate the double-stranded RNA-dependent protein kinase (PKR) pathway, and thus phosphorylate the protein translation initiation regulator eIF2α, resulting in the initiation of protein translation (Farassati et al., 2001 [↗](#)). Previous



**Figure 5.**

### The modified HSV-based constructs effectively reactivated SIV latency in vivo in chronically SIV-infected, ART-treated macaques.

(A-D) Viral load (VL) changes in plasma for each animal were monitored throughout the experiment using real-time PCR. The detection limit is 100 copies per mL plasma. The shaded area represents the duration of ART administration. (E) The VL change in plasma between pre-ART and the peak value in the rebound stage after ART discontinuation. (F) Change in total SIV DNA copies between pre-ART and viral rebound after ART discontinuation. (G) SIV integrated DNA (iDNA) copy numbers detected by Alu-PCR at different time points. (H) Change in the number of SIV Pol-specific IFN- $\gamma$ -secreting cells between pre-immunization (day 33) and post-immunization (day 70), as detected by ELISpot assay. (I) Change in the CD4<sup>+</sup> T/CD8<sup>+</sup> T ratio.



**Figure 6.**

**Pattern to illustrate the proof-of-concept strategy based on a bifunctional HSV-vectored therapeutic vaccine for HIV functional cure.**

In the present study, the modified HSV- $\Delta$ ICP34.5-based constructs effectively reactivated HIV/SIV latency by modulating the IKK $\alpha$ / $\beta$ -NF- $\kappa$ B pathway and PP1-HSF1 pathway (Shock) and simultaneously elicited antigen-specific polyfunctional CD8<sup>+</sup> T cells to eliminate cells infected with the reactivated virion (Kill). BAC: bacterial artificial chromosome; rHSV: recombinant HSV; TCR: T-cell receptor; PD1: Programmed Cell Death Protein 1; CD40L: CD40 Ligand.

studies have shown that the reactivation potential of HSV might be intertwined with NF- $\kappa$ B, Sp1, and other unknown transcription factors by ICP0, ICP4, and ICP27 (Chapman et al., 1991 [\[1\]](#), Amici et al., 2004 [\[2\]](#), Vlach and Pitha, 1992 [\[3\]](#), Mosca et al., 1987b [\[4\]](#), Golden et al., 1992 [\[5\]](#), Vlach and Pitha, 1993 [\[6\]](#), Mosca et al., 1987a [\[7\]](#)). However, the underlying mechanism by which the ICP34.5-deleted HSV construct can greatly improve the reactivation efficacy of HIV latency remains elusive. ICP34.5 is a neurotoxicity factor that can antagonize innate immune responses, including PKR, TANK binding kinase (TBK1) signaling, and Beclin1-mediated apoptosis (He et al., 1997 [\[8\]](#), Manivanh et al., 2017 [\[9\]](#), Orvedahl et al., 2007 [\[10\]](#)). ICP34.5 binds to host PP1 and mediates the dephosphorylation of eIF2 $\alpha$ , thus allowing protein synthesis and reversing the effects of PKR and host antiviral functions (He et al., 1997 [\[8\]](#), He et al., 1998 [\[11\]](#)). In this study, our findings further unveiled an interaction between ICP34.5 and HSF1, resulting in reduced HSF1 phosphorylation via recruitment of PP1 $\alpha$ . Interestingly, previous reports indicated that HSF1 could positively regulate HIV gene transcription (Rawat and Mitra, 2011 [\[12\]](#)), which is facilitated by its nuclear entry post-phosphorylation and subsequent recruitment of p300 for self-acetylation, along with binding to the HIV-1 LTR. Studies have also shown that HSF1 could further orchestrate p-TEFb recruitment to promote HIV-1 transcriptional elongation (Peng et al., 2020 [\[13\]](#), Lin et al., 2018 [\[14\]](#), Pan et al., 2016b [\[15\]](#), Pan et al., 2016a [\[16\]](#)). Under stress-induced conditions, phosphorylation triggers the formation of the HSF1 trimer, thus facilitating its nuclear entry to bind to heat shock elements (HSEs) for gene transcription regulation (Bonner et al., 2000 [\[17\]](#), Timmons et al., 2020 [\[18\]](#)). Additionally, we also demonstrated that ICP34.5 interacted with IKK $\alpha$  and IKK $\beta$ , thereby impeding NF- $\kappa$ B nuclear entry and further curbing HIV latency. Consistently, previous studies have also suggested that ICP34.5 could disrupt the NF- $\kappa$ B pathway and possibly affect the maturation of dendritic cells (Jin et al., 2011 [\[19\]](#)). Intriguingly, these findings collectively indicated that ICP34.5 might play an antagonistic role in the reactivation of HIV by HSV-1, and thus our modified HSV- $\Delta$ ICP34.5 constructs can effectively reactivate HIV/SIV latency through the release of imprisonment from ICP34.5. However, ICP34.5 overexpression had only a partial effect on the reduction of the HIV latency reactivation, indicating that HSV- $\Delta$ ICP34.5-based constructs can also reactivate HIV latency through other yet-to-be-determined mechanisms.

Another observation in this study is that the HSV-sPD1-SIVgag/SIVenv construct elicited robust and persistent SIV-specific T cell immune responses in ART-treated, chronically SIV-infected macaques. Increasing evidence has indicated that HIV-specific cytolytic T lymphocytes (CTLs) can facilitate the suppression of latent viral reservoirs, and thus, the induction of robust and persistent HIV-specific CTL responses is essential for achieving long-term disease-free and transmission-free HIV control (Collins et al., 2020 [\[20\]](#)). Featured polyfunctional CD8<sup>+</sup> T cells may contribute to HIV elite controllers or long-term non-progressors, a rare proportion of HIV-infected individuals who can spontaneously control viral replication even without ART treatment (Owen et al., 2010 [\[21\]](#), Ferre et al., 2009 [\[22\]](#), O'Connell et al., 2009 [\[23\]](#)). In addition, both our previous study and others have demonstrated that strong antigen-specific CD8<sup>+</sup> T cell immune responses, especially effector memory CD8<sup>+</sup> T cells, were associated with a lower viral load, and *in vivo* CD8<sup>+</sup> lymphocyte depletion with intravenous infusion of anti-CD8 monoclonal antibody could lead to dramatical viral rebound in these vaccinated elite macaques (Perdomo-Celis et al., 2022 [\[24\]](#), Pandrea et al., 2011 [\[25\]](#), Sun et al., 2013 [\[26\]](#), Sun et al., 2010 [\[27\]](#), Pan et al., 2018 [\[28\]](#), Sun et al., 2012 [\[29\]](#)). Notably, previous studies have shown that HSV-based vaccines expressing HIV/SIV antigen did elicit specific CD8<sup>+</sup> T cell immune responses in mice and macaques, but the magnitude was not as strong as other viral vectors (Parker et al., 2007 [\[30\]](#), Murphy et al., 2000 [\[31\]](#), Kaur et al., 2007 [\[32\]](#)). Therefore, to further improve its immunogenicity *in vivo*, some modifications were adopted to optimize the HSV-vectored vaccine. 1) The ICP47 protein, encoded by the US12 gene, can bind to the transporter associated with antigen presentation (TAP) 1/2, inhibiting the transport of viral peptides into the endoplasmic reticulum and the loading of peptides onto nascent major histocompatibility complex (MHC) class I molecules to activate CD8<sup>+</sup> T cells (Orr et al., 2005 [\[33\]](#)). Therefore, the ICP47 gene was deleted from our developed HSV vector. 2) Negatively immunoregulatory molecules, including PD1, TIM-3, and LAG-3, are usually involved in the pathogenesis of HIV infection, as well as in chronically SIV-infected macaques. Among them, PD1 upregulation can result in the exhaustion



and dysfunction of CD8<sup>+</sup> T cells (Trautmann et al., 2006 [↗](#)). In addition, PD1 expression on memory CD4<sup>+</sup> T cells might be linked with HIV latency. In this study, we modified the SIV antigen by fusing to soluble PD1 (sPD1), which can block the PD1/PDL1 pathway by competitively binding with PDL1 and thus improve HIV/SIV vaccine-induced CD8<sup>+</sup> T cell immune responses (Zhou et al., 2013 [↗](#), Wu et al., 2022 [↗](#), Pan et al., 2018 [↗](#), Xiao et al., 2014 [↗](#)).

Although promising, there are still some limitations to our study. Firstly, this is a pilot study with a relatively small numbers of RMs, and future studies with a larger number of animals can be conducted to better verify our strategy. Secondly, the HSV-sPD1-SIVgag/SIVenv vaccine resulted in delayed viral rebound and a lower peak viral load post-rebound than pre-ART treatment but did not completely suppress SIV virus rebound in this study, which may be attributed to suboptimal doses and treatment, implying that we should further optimize this regimen for eventually achieving an HIV functional cure in future studies. The current consensus on HIV/AIDS vaccines emphasizes the importance of simultaneously inducing broadly neutralizing antibodies and cellular immune responses. Therefore, we believe that incorporating the induction of broadly neutralizing antibodies into our future optimizing approaches may lead to better therapeutic outcomes. Taken together, these findings demonstrated that our modified HSV-ΔICP34.5-based constructs potentially reactivated HIV/SIV latency by modulating the IKKα/β-NF-κB pathway and PP1-HSF1 pathway, and thereby we developed a proof-of-concept strategy based on a bifunctional HSV-vectored therapeutic vaccine, which can provide insights into the rational design of novel strategies for pursuing an HIV functional cure.

## Methods

### Mouse ethics statement and vaccination

Female BALB/c mice, aged six to eight weeks, were procured from the Experimental Animal Center of Sun Yat-sen University. A total of twenty-five mice were randomly divided into five groups to evaluate the immunogenicity of the recombinant HSV-vector-based SIV vaccine. During the initial week, each mouse was subcutaneously administered a vaccination of  $1 \times 10^6$  PFU of the respective recombinant HSV-vector vaccines (HSV-empty, HSV-sPD1, HSV-SIVgag, HSV-sPD1-SIVgag, HSV-SIVenv). Following a two-week interval, a booster vaccination was administered via the subcutaneous route using  $2 \times 10^6$  PFU of recombinant HSV-vector vaccines. The subsequent assessment of the immune response involved ELISpot and intracellular cytokine staining (ICS) assays in accordance with the vaccination schedule.

### Ethics statement and vaccination of macaques

A total of 9 chronically SIV<sub>mac239</sub>-infected rhesus macaques (RMs) were included in this study. All RMs received ART treatment (FTC/6.7 mg/kg/once daily, PMPA/10 mg/kg/once daily) for a duration of 33 days. The RMs were allocated into three groups based on their baseline plasma SIV viral loads (VL): ART + saline (n=3) as the control group, ART+ HSV-empty (n=3) as the sham group, and ART+ HSV-sPD1-SIVgag/SIVenv (n=3) as the vaccinated group. On day 0, all RMs received ART treatment. Once the VL dropped below the limit of detection ( $1 \times 10^2$  copies/mL), the vaccinated group and sham group were primed with a subcutaneous vaccination of  $1 \times 10^7$  PFU HSV-sPD1-SIVgag/SIVenv or HSV-empty, respectively, while the control group received an injection of 0.9% saline. After three weeks of the prime vaccination, the vaccinated group and sham group RMs received a subcutaneous booster vaccination of  $5 \times 10^7$  PFU HSV-sPD1-SIVgag/SIVenv or HSV-empty, respectively. After two weeks of booster vaccination, the ART treatment for all RMs was interrupted. The immune response was evaluated by IFN-γ ELISpot and ICS assays. Plasma viral load was regularly monitored throughout the studied period.

## Peptide pools

The peptide pools, encompassing the complete sequences of SIV Gag, Env, and Pol proteins, comprising 15 amino acids with 11 overlaps, were generously provided by the HIV Reagent Program, National Institutes of Health (NIH), USA. Gag pools comprise 125 peptides, while Env and Pol pools are subdivided into Env1 (109 peptides) and Env2 (109 peptides) pools, as well as Pol1 (131 peptides) and Pol2 (132 peptides) pools. These peptide pools were dissolved in dimethyl sulfoxide (DMSO, Sigma) at a concentration of 0.4 mg/ peptide/mL for subsequent immunological assays.

## Construction of recombinant HSV

For the generation of recombinant HSV-vector-based vaccines, we implemented modifications at the ICP34.5 loci through homologous recombination within the BAC-HSV-1 system. Specifically, double copies of the ICP34.5 and ICP47 genes were either deleted or inserted into the respective counterparts: sPD1, SIVgag, sPD1-SIVgag, and SIVenv genes. The sPD1-SIVgag gene was created by fusing the N-terminal region of mouse soluble PD1 (sPD1) with the C-terminal segment of SIVgag, connected by a GGGSGGG linker, which was engineered through overlapping PCR.

## Plaque Assay

Vero cells were seeded at a density of  $2 \times 10^5$  cells per well in a 12-well plate and cultured for 24 h. Virus samples were serially diluted and then added to the wells. After 2 h of incubation at 37°C with 5% CO<sub>2</sub>, the supernatant was aspirated, and the cells were washed with PBS. Then, a medium containing 1% FBS, 1% low-melting-point agarose, and 1% penicillin-streptomycin was added to each well. After 3 days of incubation, cells were fixed with 4% paraformaldehyde at room temperature for 2 h and stained with 1% crystal violet for 10 min. Plaques were visualized and counted, and virus titers were calculated as plaque-forming units per milliliter (PFU/mL).

## IFN-γ ELISpot assay

The IFN-γ-ELISpot assay was conducted in accordance with our previous study (Sun et al., 2010 [\[10\]](#)). In the mouse experiment,  $2.5 \times 10^5$  freshly isolated mouse spleen lymphocytes were stimulated with SIV Gag, Env1, and Env2 peptide pools. In the monkey experiment,  $2 \times 10^5$  peripheral blood mononuclear cells (PBMCs) were seeded and stimulated with the SIV Gag, Env, and Pol peptide pools. Mock stimulation utilized DMSO (Sigma), while concanavalin A (ConA, 10 μg/mL) was employed as a positive control. Spot quantification was performed using an ELISpot reader (Mabtech), and peptide-specific spot counts were determined by subtracting the spots from mock stimulation.

## Intracellular cytokine staining (ICS)

The ICS assay was performed as described previously (Wu et al., 2022 [\[11\]](#)). In the mouse experiment,  $2 \times 10^6$  freshly isolated mouse spleen lymphocytes were stimulated with SIV Gag, Env1, and Env2 peptide pools. For the monkey experiment,  $2 \times 10^6$  PBMCs were seeded and stimulated with the SIV Gag, Env, and Pol peptide pools. DMSO and PMA (Thermo Scientific) plus ionomycin were used as the mock and positive controls, respectively. Data analysis was performed using FlowJo software (version 10.8.1), and the antibodies used are listed in Table S3. The frequencies of cytokines produced from peptide-specific cells were analyzed by subtracting mock stimulation.

## SIV viral RNA and DNA copy assays and absolute T cell count

Absolute T cell count and the levels of plasma mRNA and SIV total DNA were quantified as described previously (Wu et al., 2022 [\[11\]](#)). Plasma viral RNA copy numbers were determined via SYBR green-based real-time quantitative PCR (Takara), using SIV gag-specific primers (Table S4). Viral RNA copy numbers were calculated based on the standard curve established using SIV<sub>mac239</sub>.

gag standards. The lower limit of detection for this assay was 100 copies/mL of plasma. Total cellular DNA was extracted from approximately 0.5 to 5 million cells using a QIAamp DNA Blood Minikit (Qiagen). PCR assays were performed with 200 ng samples of DNA, and SIV viral DNA was quantified using a pair of primers targeting a conserved region of the SIV gag gene, as previously described. Quantitation was performed by comparing the results to the standard curve of SIV gag copies. Integrated SIVmac239 DNA was measured using nested PCR with two sets of primers. The first round used SIVgag-reverse and Alu-forward primers, while the second round used primers specific to a conserved SIVgag region. Quantification was compared to a standard of CEMss/pWPXLD-rc cell genome as our previously described (Yang et al., 2019 [DOI](#)).

## Plasmids, cells, and viruses

pVAX-Myc-PDL1, pVAX-Flag-ICP34.5, pVAX-Myc-HSF1, pVAX-HA-PP1α: full-length mouse PDL1, full-length HSV-1 ICP34.5, and full-length human HSF1 and human PP1α with N or C-terminal Tag were cloned into the pVAX vector. pVAX-empty, pcDNA3.1-IKKα, and pcDNA3.1-IKKβ plasmids were stored in our laboratory. pVAX-empty was used as a mock transfection in our study.

293T cells (from the embryonic kidney of a female human fetus), Vero cells (from the kidney of a female normal adult African green monkey), and Hela cells (from the cervical cancer cells of an African American woman) were cultured in complete Dulbecco's modified Eagle's medium (DMEM, Gibco) containing 10% fetal bovine serum (FBS, Gibco) and 1% penicillin/streptomycin (Gibco) at 37°C in an atmosphere of 5% CO<sub>2</sub>. The J-Lat 10.6 cells (Jurkat cells contain the HIV-1 full-length genome whose Env was frameshifted and inserted with GFP in place of Nef) and the HIV-1 latently infected CD4<sup>+</sup> CEM cells ACH-2 (A3.01 cell integrated HIV-1 proviral DNA) were cultured in complete RPMI640 (Gibco) containing 10% FBS and 1% penicillin/streptomycin at 37°C in an atmosphere of 5% CO<sub>2</sub>. J-Lat 10.6-ICP34.5 cells were constructed in our laboratory.

The HSV-1 (McKrae) strains were stored in our laboratory. HSV-GFP (17 strain) was rescued from pBAC-GFP-HSV in Vero cells. HSV-ΔICP34.5 (17 strain) was rescued from pBAC-GFP-HSV-ΔICP34.5 (with deletion of double copies of ICP34.5 gene) in Vero cells.

## RT-qPCR

RT-qPCR was performed as described in our previous study (Zhao et al., 2022 [DOI](#)). Data were normalized to β-actin. The primer sequences are listed in Supplemental Table 4. Fold changes in the threshold cycle (Ct) values were calculated using the 2<sup>-ΔΔCt</sup> method.

## Chromatin immunoprecipitation (ChIP) assay

ChIP analysis was conducted using the SimpleChIP Enzymatic Chromatin IP kit (Agarose Beads) (CST), following the manufacturer's instructions. Quantitative real-time PCR was employed for detecting the LTR sequence. The sequences of primers used in the LTR ChIP are listed in Supplemental Table 4. % Input =  $2\% \times 2^{[(ct)2\% \text{ input sample} (ct) \text{ IP sample}]}$ .

## Co-immunoprecipitation (Co-IP)

Cells were harvested and subjected to Co-IP assay, following the protocol outlined in our previous study (Zhao et al., 2022 [DOI](#)).

## Protein extraction and western blot

Nuclear and cytoplasmic proteins were extracted by kits following the manufacturer's instructions (Beyotime). The Western blotting assay was performed as previously described (Zhao et al., 2022 [DOI](#)).

## Statistical analysis

GraphPad Prism 8.0 (GraphPad Software, San Diego, California) was used for statistical analysis. For intragroup direct comparisons, Student's unpaired two-tailed *t* test was performed to analyze significant differences. For comparisons of multiple groups, one-way ANOVAs were performed. Significance levels are indicated as \**P* < 0.05, \*\* *P* < 0.01, \*\*\* *P* < 0.001, \*\*\*\* *P* < 0.0001.

## Study approval

Mice experiment was approved by the Laboratory Animal Ethics Committee guidelines at Sun Yat-sen University (approval number: SYSU-IACUC-2021-000185). Chinese rhesus macaques (*Macaca mulatta*) were housed at the Landau Animal Experimental Center, Guangdong Landau Biotechnology Co., Ltd. (approval number: N2021101). The primary PBMC samples were isolated from the chronically HIV-1-infected participants who were recruited from Guangzhou Eighth People's Hospital (Guangzhou, Guangdong, China), which was approved by the Medical Ethics Review Board of the School of Public Health (Shenzhen), Sun Yat-sen University (2022-037). All participants provided written informed consent, agreeing to participate in this study. The enrollment criteria for HIV-1-infected individuals included sustained suppression of plasma HIV-1 viremia under ART, with undetectable plasma HIV-1 RNA levels (less than 50 copies/mL) for at least 12 months and a CD4<sup>+</sup> T cell count more than 350 cells/μL).

## Data Availability Statement

Source data contain the numerical data used to generate the figures.

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## Additional information

### Competing interests

The authors have declared that no conflict of interest exists.

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## Author Contributions

All authors were involved in drafting or critically revising the manuscript for important intellectual content, and all authors approved the final version for publication. CS conceived the project and provided the main funding. CS and LC designed the experiments and reviewed the manuscript. ZW, PL, YY, and CW performed most of the experiments and analyzed the data. ZW, PL, YY, CW, ML, HW, MS, YH, and MC established the animal models and performed animal experiments. ZW and CS interpreted the data and wrote the manuscript.



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## Editors

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### Reviewer #1 (Public review):

Summary:

The authors constructed a novel HSV-based therapeutic vaccine to cure SIV in a primate model. The novel HSV vector is deleted for ICP34.5. Evidence is given that this protein blocks HIV reactivation by interference with the NFkappaB pathway. The deleted construct supposedly would reactivate SIV from latency. The SIV genes carried by the vector ought to elicit a strong immune response. Together the HSV vector would elicit a shock and kill effect. This is tested in a primate model.

Strengths and weaknesses:

(1) Deleting ICP34.5 from the HSV construct has a very strong effect on HIV reactivation. The mechanism underlying increased activation by deleting ICP34.5 is only partially explored. Overexpression of ICP34.5 has a much smaller effect (reduction in reactivation) than deletion of ICP34.5 (strong activation); this is acknowledged by the authors that no full mechanistic explanation can be given at this moment.

(2) No toxicity data are given for deleting ICP34.5. How specific is the effect for HIV reactivation? A RNA seq analysis is required to show the effect on cellular genes.

A RNA seq analysis was done in the revised manuscript comparing the effect of HSV-1 and deleted vector in J-LAT cells (Fig S5). More than 2000 genes are upregulated after transduction with the modified vector in comparison with the WT vector. Hence, the specificity of upregulation of SIV genes is questioned. Authors do NOT comment on these findings. In my view it questions the utility of this approach.

(3) The primate groups are too small and the results too variable to make averages. In Fig 5, the group with ART and saline has two slow rebounders. It is not correct to average those with the single quick rebounder. Here the interpretation is NOT supported by the data.

Although authors provided some promising SIV DNA data, no additional animals were added. Groups of 3 animals are too small to make any conclusion, especially since the huge variability in response. The average numbers out of 3 are still presented in the paper, which is not proper science.



No data are given of the effect of the deletion in primates. Now the deleted construct is compared with an empty vector containing no SIV genes. Authors provide new data in Fig S2 on the comparison of WT and modified vector in cells from PLWH, but data are not that convincing. A significant difference in reactivation is seen for LTR in only 2/4 donors and in Gag in 3/4 donors. (Additional question what is meaning of LTR mRNA, do authors relate to genomic RNA??)

#### Discussion

HSV vectors are mainly used in cancer treatment partially due to induced inflammation. Whether these are suitable to cure PLWH without major symptoms is a bit questionable to me and should at least be argued for.

The RNA seq data add on to this worry and should at least be discussed.

<https://doi.org/10.7554/eLife.95964.2.sa2>

#### Reviewer #2 (Public review):

##### Summary:

In this article Wen et. al., describe the development of a 'proof-of-concept' bi-functional vector based out of HSV-deltaICP-34.5's ability to purge latent HIV-1 and SIV genomes from cells. They show that co-infection of latent J-lat T-cell lines with a HSV-deltaICP-34.5 vector can reactivate HIV-1 from a latent state. Over- or stable expression of ICP 34.5 ORF in these cells can arrest latent HIV-1 genomes from transcription, even in the presence of latency reversal agents. ICP34.5 can co-IP with- and de-phosphorylate IKK $\alpha$ /b to block its interaction with NF- $\kappa$ B transcription factor. Additionally, ICP34.5 can interact with HSF1 which was identified by mass-spec. Thus, the authors propose that the latency reversal effect of HSV-deltaICP-34.5 in co-infected JLat cells is due to modulatory effects on the IKK $\alpha$ /b-NF- $\kappa$ B and PP1-HSF-1 pathway.

Next the authors cleverly construct a bifunctional HSV based vector with deleted ICP34.5 and 47 ORFs to purge latency and avoid immunological refluxes, and additionally expand the application of this construct as a vaccine by introducing SIV genes. They use this 'vaccine' in mouse models and show the expected SIV-immune responses. Experiments in rhesus macaques (RM), further elicit potential for their approach to reactivate SIV genomes and at the same time block their replication by antibodies. What was interesting in the SIV experiments is that the dual-functional vector vaccine containing SPD1- and SIV Gag/Env ORFs effectively delayed SIV rebound in RMs and in some cases almost neutralized viral DNA copy detection in serum. Very promising indeed, however there are some questions I wish the authors explored to answer, detailed below.

Overall, this is an elegant and timely work demonstrating the feasibility of reducing virus rebound in animals, and potentially expand to clinical studies. The work was well written, and sections were clearly discussed.

##### Strengths:

The work is well designed, rationale explained and written very clearly for lay readers. Claims are adequately supported by evidence and well designed experiments including controls.

##### Weaknesses:

(1) It looks like ICP0 is also involved in latency reversal effects. More follow-up work will be required to test if this is in fact true.

(2) It is difficult to estimate the depletion of the latent viral reservoir. The authors have tried to address this issue. A more convincing argument to this reviewer will be data to demonstrate that after the bi-functional vaccine, the animals show overall reduction in the number of circulating latent cells. The feasibility to obtain such a result is not clearly demonstrated.

(3) The authors state that the reduced virus rebound detected following bi-functional vaccine delivery is due to latent genomes becoming activated and steady-state neutralization of these viruses by antibody response. This needs to be demonstrated. Perhaps cell-culture experiments from specimen taken from animals might help address this issue. In lab cultures one could create environments without antibody responses, under these conditions one would expect higher level of viral loads being released in response to the vaccine in question.

<https://doi.org/10.7554/eLife.95964.2.sa1>

Author response:

The following is the authors' response to the original reviews.

**Reviewer #1 (Public Review):**

*Summary:*

*The authors constructed a novel HSV-based therapeutic vaccine to cure SIV in a primate model. The novel HSV vector is deleted for ICP34.5. Evidence is given that this protein blocks HIV reactivation by interference with the NF- $\kappa$ B pathway. The deleted construct supposedly would reactivate SIV from latency. The SIV genes carried by the vector ought to elicit a strong immune response. Together the HSV vector would elicit a shock and kill effect. This is tested in a primate model.*

Thank you for your kind comments and suggestions, which are very helpful in improving our manuscript. We have carefully revised our manuscript and performed additional experiments accordingly, and we now think this version has been substantially improved for your reconsideration.

*Strengths and weaknesses:*

*(1) Deleting ICP34.5 from the HSV construct has a very strong effect on HIV reactivation. Why is no eGFP readout given in Figure 1C as for WT HSV? The mechanism underlying increased activation by deleting ICP34.5 is only partially explored. Overexpression of ICP34.5 has a much smaller effect (reduction in reactivation) than deletion of ICP34.5 (strong activation); so the story seems incomplete.*

Thank you for your careful review and kind reminder.

(1) We are sorry for the misunderstanding of Figure 1C. In the experiment of Figure 1C, we used an HSV-1 17 strain containing GFP (HSV-GFP) and HSV-DICP34.5 (recombinant HSV-1 17 strain with ICP34.5 deletion based on HSV-GFP) to reactivate the HIV latency cell line (J-Lat 10.6 cell). Since detecting GFP cannot distinguish between HSV infection and HIV reactivation, we assessed the reactivation by measuring the mRNA levels of HIV LTR upon stimulation with either HSV-GFP or HSV- $\Delta$ ICP34.5. Actually, in Figure 1B, we had verified the reactivation efficacy by infecting J-Lat 10.6 cells with the HSV-1 17 strain containing GFP

(HSV-GFP) and found significant upregulation of mRNA levels of HIV-1 LTR, Tat, Gag, Vif, and Vpr. We have adjusted the corresponding descriptions accordingly in the revised manuscript.

(2) We agree with your insightful mention that the mechanism underlying increased activation by HSV-ΔICP34.5 is worthy to be further explored in the future study. In this study, we found that ICP34.5 play an antagonistic role with the reactivation of HIV latency by HSV-1 mainly through the modulation of host NF-κB and HSF1 pathways, while HSV-1 (especially HSV-ΔICP34.5) might reactivate HIV latency through NF-κB, HSF1, and other yet-to-be-determined mechanisms. Thus, ICP34.5 overexpression can only a partial effect on the reduction of the HIV latency reactivation by HSV-1. We have mentioned this issue in the revised “Discussion section”. *“Intriguingly, these findings collectively indicated that ICP34.5 might play an antagonistic role in the reactivation of HIV by HSV-1, and thus our modified HSV-DICP34.5 constructs can effectively reactivate HIV/SIV latency through the release of imprisonment from ICP34.5. However, ICP34.5 overexpression had only a partial effect on the reduction of the HIV latency reactivation, indicating that HSV-DICP34.5-based constructs can also reactivate HIV latency through other yet-to-be-determined mechanisms.”* (Lines 334 to 340).

*(2) No toxicity data are given for deleting ICP34.5. How specific is the effect for HIV reactivation? An RNA seq analysis is required to show the effect on cellular genes.*

Thank you for your questions and suggestions.

(1) It's well known that ICP34.5 is a neurotoxicity factor that can antagonize host immune responses, and previous studies (in gene therapy and oncolytic virotherapy) have shown that the safety of recombinant HSV-based vector can be improved by deleting ICP34.5. In this study, we also found that HSV-DICP34.5 exhibited lower virulence and replication ability than its parental strain (HSV-GFP) (Figure 1D, Figure S1). In addition, HSV-DICP34.5 induced a lower level of inflammatory cytokines (including IL-6, IL-1β, and TNF-α) in primary CD4+ T cells from PLWH compared to HSV-GFP stimulation, likely due to its lower virulence and replication ability (Figure 1I-K). In addition, the CD4+ /CD8+ T cell ratio (Figure 5I) and body weight (Figure S9) after treatment were effectively ameliorated in the SIV-infected macaques of the ART+HSV-DICP34.5-sPD1-SIVgag/SIVenv group. Our data also demonstrated that there was no significant effect on the cell composition of peripheral blood in the SIV-infected macaques of ART+HSV-sPD1-SIVgag/SIVenv group (Figure S10). Thus, these data suggest the safety of HSV-DICP34.5 in PLWH might be tolerable. We have added the corresponding description in the revised manuscript.

(2) In our study, we found both adenovirus and vaccinia virus cannot reactivate HIV latency (Figure S3). In addition, the deletion of ICP0 gene from HSV-1 diminished the reactivation effect of HIV latency by HSV-1 (Figure S4). Thus, these data suggested the reactivation of HIV latency by HSV-1 might be virus-specific. Of course, this might be further investigated in future studies. We have added the corresponding description in the revised manuscript.

(3) To explore the mechanism of reactivating viral latency by HSV-DICP34.5-based constructs, we performed RNA-seq analysis (Figure S5). We have added the corresponding description accordingly in the revised manuscript.

*(3) The primate groups are too small and the results to variable to make averages. In Figure 5, the group with ART and saline has two slow rebounders. It is not correct to average those with a single quick rebounder. Here the interpretation is NOT supported by the data.*

We agree with you that this is a pilot study with limited numbers of rhesus macaques. Although the number of macaques was relatively limited, these nine macaques were

distributed evenly based on the background level of age, sex, weight, CD4 count, and viral load (VL) (Table S2). All SIV-infected macaques used in this study had a long history of SIV infection and had several courses of ART therapy, which mimics treatment of chronic HIV-1 infection in humans. These macaques were infected with SIVmac239 for more than 5 years, and highly pathogenic SIV-infected macaques have been well-validated as a stringent model to recapitulate HIV-1 pathogenesis and persistence during ART therapy in humans. Indeed, in our Chinese rhesus model, ART treatment effectively suppressed SIV infection to undetectable levels in plasma, and upon ART discontinuation, virus rapidly rebounded, which is very similar with that in ART-treated HIV patients. We think the results of this pilot study were very promising for further studies which will be expanded the scale of animals and then to preclinical and clinical study in our next projects. Thank you for your understanding.

As for your question regarding “the two animals with low VL and slow rebound”, our explanation is following: As mentioned above, these macaques were distributed evenly based on the background level of CD4 count and VL (Table S2), and then there were different change of viral load and viral rebound in different groups. Thus, we think these data can support our interpretation. Moreover, our conclusion can also be supported from at least three evidences.

(1) The VL in the ART+saline group promptly rebounded after ART discontinuation, with an average 8.63-fold increase in the rebounded peak VL compared with the pre-ART VL (Figure 5A, D and E). However, plasma VL in the ART+HSV-sPD1-SIVgag/SIVenv group exhibited a delayed rebound interval (Figure 5B-D).

(2) There was a lower rebounded peak VL than pre-ART VL in the ART+HSV-sPD1-SIVgag/SIVenv group (average 12.20-fold decrease), while a higher rebounded peak VL than pre-ART VL in the ART+HSV-empty group (average 2.74-fold increase) (Figure 5E).

(3) We found significant suppression of total SIV DNA and integrated SIV DNA provirus in the ART+HSV-sPD1-SIVgag/SIVenv group. However, the copies of the SIV DNA provirus were significantly improved in the ART+HSV-empty group and ART+saline group (Figure 5F-G).

Thank you for your understanding.

#### *Discussion*

*HSV vectors are mainly used in cancer treatment partially due to induced inflammation. Whether these are suitable to cure PLWH without major symptoms is a bit questionable to me and should at least be argued for.*

Thank you for your kind question comment and question. We confirmed the enhanced reactivation of HIV latency by HSV-ΔICP34.5 in primary CD4<sup>+</sup> T cells from people living with HIV (PLWH) (Figure S2). As mentioned above, previous studies have shown that the safety of recombinant HSV-based vector can be improved by deleting ICP34.5. In this study, we also found that HSV-DICP34.5 exhibited lower virulence and replication ability than its parental strain (HSV-GFP) (Figure 1D, Figure S1). In addition, HSV-DICP34.5 induced a lower level of inflammatory cytokines (including IL-6, IL-1β, and TNF-α) in primary CD4<sup>+</sup> T cells from PLWH compared to HSV-GFP stimulation, likely due to its lower virulence and replication ability (Figure 1I-K). In addition, the CD4<sup>+</sup>/CD8<sup>+</sup> T cell ratio (Figure 5I) and body weight (Figure S9) after treatment were effectively ameliorated in the SIV-infected macaques of the ART+HSV-DICP34.5-sPD1-SIVgag/SIVenv group. Our data also demonstrated that there was no significant effect on the cell composition of peripheral blood in the SIV-infected macaques of ART+HSV-sPD1-SIVgag/SIVenv group (Figure S10). Thus, these data suggest the safety of HSV-DICP34.5 in PLWH might be tolerable. We have added the corresponding description in the revised manuscript.

## Reviewer #2 (Public Review):

### Summary:

*In this article, Wen et. al. describe the development of a 'proof-of-concept' bi-functional vector based on HSV-deltaICP-34.5's ability to purge latent HIV-1 and SIV genomes from cells. They show that co-infection of latent J-lat T-cell lines with an HSV-deltaICP-34.5 vector can reactivate HIV-1 from a latent state. Over- or stable expression of ICP 34.5 ORF in these cells can arrest latent HIV-1 genomes from transcription, even in the presence of latency reversal agents. ICP34.5 can co-IP with- and de-phosphorylate IKK $\alpha$ /b to block its interaction with NF-k/B transcription factor. Additionally, ICP34.5 can interact with HSF1 which was identified by mass-spec. Thus, the authors propose that the latency reversal effect of HSV-deltaICP-34.5 in co-infected JLat cells is due to modulatory effects on the IKK $\alpha$ /b-NF-kB and PP1-HSF-1 pathway.*

*Next, the authors cleverly construct a bifunctional HSV-based vector with deleted ICP34.5 and 47 ORFs to purge latency and avoid immunological reflexes, and additionally, expand the application of this construct as a vaccine by introducing SIV genes. They use this 'vaccine' in mouse models and show the expected SIV-immune responses. Experiments in rhesus macaques (RM), further elicit the potential for their approach to reactivate SIV genomes and at the same time block their replication by antibodies. What was interesting in the SIV experiments is that the dual-functional vector vaccine containing sPD1- and SIV Gag/Env ORFs effectively delayed SIV rebound in RMs and in some cases almost neutralized viral DNA copy detection in serum. Very promising indeed, however, there are some questions I wish the authors had explored to get answers to, detailed below.*

*Overall, this is an elegant and timely work demonstrating the feasibility of reducing virus rebound in animals, with the potential to expand to clinical studies. The work was well-written, and sections were clearly discussed.*

### Strengths:

*The work is well designed, rationale explained, and written very clearly for lay readers. Claims are adequately supported by evidence and well-designed experiments including controls.*

Thank you for your nice comments regarding our work.

### Weaknesses:

*(1) While the mechanism of ICP34.5 interaction and modulation of the NF-kB and HSF1 pathways are shown, this only proves ICP34.5 interactions but does not give away the mechanism of how the HSV-deltaICP-34.5 vector purges HIV-1 latency. What other components of the vector are required for latency reversal? Perhaps serial deletion experiments of the other ORFs in the HSV-deltaICP-34.5 vector might be revealing.*

Thank you for your valuable suggestion. In fact, we are currently further exploring some potential viral genes of HSV-1 that might play a role in the reactivation of HIV latency. We have found that the deletion of ICP0 gene from HSV-1 diminished the reactivation effect of HIV latency by HSV-1 (Figure S4), showing that ICP0 might play a vital role for the reactivation. Of course, this might be further investigated in future studies. We have added the corresponding description in the revised manuscript.

*(2) The efficacy of the HSV vaccine vectors was evaluated in Rhesus Macaque model animals. Animals were chronically infected with SIV (a parent of HIV), treated with ART, challenged with bi-functional HSV vaccine or controls, and discontinued treatment, and the resulting virus burden and immune responses were monitored. The animals showed SIV Gag and Env-specific immune responses, and delayed virus rebound (however rebound is still there), and below-detection viral DNA copies. What would make a more convincing argument to this reviewer will be data to demonstrate that after the bi-functional vaccine, the animals show overall reduction in the number of circulating latent cells. The feasibility of obtaining such a result is not clearly demonstrated.*

Thank you for your valuable mention. We have now provided more data about this issue. We found significant suppression of total SIV DNA and integrated SIV DNA provirus in the ART+HSV-sPD1-SIVgag/SIVenv group. However, the copies of the SIV DNA provirus were significantly improved in the ART+HSV-empty group and ART+saline group (Figure 5F-G). We have added the corresponding description in the revised manuscript.

*(3) The authors state that the reduced virus rebound detected following bi-functional vaccine delivery is due to latent genomes becoming activated and steady-state neutralization of these viruses by antibody response. This needs to be demonstrated. Perhaps cell-culture experiments from specimens taken from animals might help address this issue. In lab cultures one could create environments without antibody responses, under these conditions one would expect a higher level of viral loads to be released in response to the vaccine in question.*

Thanks for your kind mention and suggestion. We performed the following cell experiment to address this issue. Primary CD4<sup>+</sup> T cells from people living with HIV (PLWH) were isolated, and then infected with HSV or HSV-ΔICP34.5 constructs. As expected, we confirmed the enhanced reactivation of HIV latency by HSV-ΔICP34.5 (Figure S2). Thank you.

*(4) How do the authors imagine neutralizing HIV-1 envelope epitopes by a similar strategy? A discussion of this point may also help.*

Thank you for your kind comment. We have added the corresponding discussion in the revised manuscript. "The current consensus on HIV/AIDS vaccines emphasizes the importance of simultaneously inducing broadly neutralizing antibodies and cellular immune responses. Therefore, we believe that incorporating the induction of broadly neutralizing antibodies into our future optimizing approaches may lead to better therapeutic outcomes." (Lines 384 to 388)

*(5) I thought the empty HSV-vector control also elicited somewhat delayed kinetics in virus rebound and neutralization, can the authors comment on why this is the case?*

Thank you for your careful review and mention. We agree with you that the HSV-1 empty vector does exhibit somewhat a delayed rebound. We think the possible reason is: Although the empty HSV-vector cannot elicit SIV-specific CTL responses, it effectively activates the latent SIV reservoirs, and then these activated virions can be partially killed by ART drugs. Therefore, even without carrying HIV/SIV antigens, somewhat delayed kinetics in virus rebound may be observed. Thank you.

#### **Reviewer #1 (Recommendations For The Authors):**

*(1) The authors should provide toxicity data for HSV transduction after deleting ICP34.5 and provide an explanation of why overexpression of ICP34.5 has such a small effect.*



Thank you for your questions and suggestions. As mentioned above, we now provided data for the safety of HSV-DICP34.5-based constructs.

(1) It's well known that ICP34.5 is a neurotoxicity factor that can antagonize host immune responses, and previous studies (in gene therapy and oncolytic virotherapy) have shown that the safety of recombinant HSV-based vector can be improved by deleting ICP34.5. In this study, we also found that HSV-DICP34.5 exhibited lower virulence and replication ability than its parental strain (HSV-GFP) (Figure 1D, Figure S1). In addition, HSV-DICP34.5 induced a lower level of inflammatory cytokines (including IL-6, IL-1 $\beta$ , and TNF- $\alpha$ ) in primary CD4<sup>+</sup> T cells from PLWH compared to HSV-GFP stimulation, likely due to its lower virulence and replication ability (Figure 1I-K). In addition, the CD4<sup>+</sup> /CD8<sup>+</sup> T cell ratio (Figure 5I) and body weight (Figure S9) after treatment were effectively ameliorated in the SIV-infected macaques of the ART+HSV-DICP34.5-sPD1-SIVgag/SIVenv group. Our data also demonstrated that there was no significant effect on the cell composition of peripheral blood in the SIV-infected macaques of ART+HSV-sPD1-SIVgag/SIVenv group (Figure S10). Thus, these data suggest the safety of HSV-DICP34.5 in PLWH might be tolerable. We have added the corresponding description in the revised manuscript.

(2) We agree with your insightful mention that the mechanism underlying increased activation by HSV- $\Delta$ ICP34.5 is worthy to be further explored in the future study. In this study, we found that ICP34.5 play an antagonistic role with the reactivation of HIV latency by HSV-1 mainly through the modulation of host NF- $\kappa$ B and HSF1 pathways, while HSV-1 (especially HSV- $\Delta$ ICP34.5) might reactivate HIV latency through NF- $\kappa$ B, HSF1, and other yet-to-be-determined mechanisms. Thus, ICP34.5 overexpression can only a partial effect on the reduction of the HIV latency reactivation by HSV-1. We have mentioned this issue in the revised "Discussion section". "Intriguingly, these findings collectively indicated that ICP34.5 might play an antagonistic role in the reactivation of HIV by HSV-1, and thus our modified HSV-DICP34.5 constructs can effectively reactivate HIV/SIV latency through the release of imprisonment from ICP34.5. However, ICP34.5 overexpression had only a partial effect on the reduction of the HIV latency reactivation, indicating that HSV-DICP34.5-based constructs can also reactivate HIV latency through other yet-to-be-determined mechanisms." (Lines 334 to 340).

*(2) How specific is the effect for HIV reactivation? An RNA seq analysis is required to show the effect on cellular genes.*

Thank you for your questions and suggestions.

(1) In our study, we found both adenovirus and vaccinia virus cannot reactivate HIV latency (Figure S3). In addition, the deletion of ICP0 gene from HSV-1 diminished the reactivation effect of HIV latency by HSV-1 (Figure S4). Thus, these data suggested the reactivation of HIV latency by HSV-1 might be virus-specific. Of course, this might be further investigated in future studies. We have added the corresponding description in the revised manuscript.

(2) To explore the mechanism of reactivating viral latency by HSV-DICP34.5-based constructs, we performed RNA-seq analysis (Figure S5). Results showed that there were numerous differentially expressed genes (DEGs) in response to HSV- $\Delta$ ICP34.5 infection. Among them, 2288 genes were upregulated, and 611 genes were downregulated. GO analysis showed the enrichment of these DEGs in cellular cycle, cellular development, and cellular proliferation, and KEGG enrichment analysis indicated the enrichment in pathways such as cellular cycle and cytokine-cytokine receptor interaction. We have added the corresponding description accordingly in the revised manuscript.

*(3) A comparison in primates has to be given for constructs with or without ICP34.5 to validate cell culture data (what is an empty vector?)*

Thank you for your reminder. In the revised manuscript, we performed the following cell experiment to address this issue. Primary CD4<sup>+</sup> T cells from people living with HIV (PLWH) were isolated, and then infected with HSV or HSV-ΔICP34.5 constructs. As expected, we confirmed the enhanced reactivation of HIV latency by HSV-ΔICP34.5 (Figure S2). Thank you.

| (4) *Legends should be improved in writing and content.*

Thank you for your kind mention. In the revised version, we have improved both the manuscript content and the legends of all Figures have been carefully revised in writing and content. Thank you.

| (5) *The primate groups should be enlarged before any reliable conclusions can be made. Inflammatory/tox data should be provided.*

Thank you for your question.

(1) As mentioned above, we agree with you that this is a pilot study with limited numbers of rhesus macaques. Although the number of macaques was relatively limited, these nine macaques were distributed evenly based on the background level of age, sex, weight, CD4 count, and viral load (VL) (Table S2). All SIV-infected macaques used in this study had a long history of SIV infection and had several courses of ART therapy, which mimics treatment of chronic HIV-1 infection in humans. These macaques were infected with SIVmac239 for more than 5 years, and highly pathogenic SIV-infected macaques have been well-validated as a stringent model to recapitulate HIV-1 pathogenesis and persistence during ART therapy in humans. Indeed, in our Chinese rhesus model, ART treatment effectively suppressed SIV infection to undetectable levels in plasma, and upon ART discontinuation, virus rapidly rebounded, which is very similar with that in ART-treated HIV patients. We think the results of this pilot study were very promising for further studies which will be expanded the scale of animals and then to preclinical and clinical study in our next projects. Thank you for your understanding.

(2) As well known, ICP34.5 is a neurotoxicity factor that can antagonize host immune responses, and previous studies have shown that the safety of recombinant HSV-based vector can be improved by deleting ICP34.5. In this study, we also found that HSV-DICP34.5 exhibited lower virulence and replication ability than its parental strain (HSV-GFP) (Figure 1D, Figure S1). In addition, HSV-DICP34.5 induced a lower level of inflammatory cytokines (including IL-6, IL-1 $\beta$ , and TNF- $\alpha$ ) in primary CD4<sup>+</sup> T cells from PLWH compared to HSV-GFP stimulation, likely due to its lower virulence and replication ability (Figure 1I-K). In addition, the CD4<sup>+</sup>/CD8<sup>+</sup> T cell ratio (Figure 5I) and body weight (Figure S9) after treatment were effectively ameliorated in the SIV-infected macaques of the ART+HSV-DICP34.5-sPD1-SIVgag/SIVenv group. Our data also demonstrated that there was no significant effect on the cell composition of peripheral blood in the SIV-infected macaques of ART+HSV-sPD1-SIVgag/SIVenv group (Figure S10). Thus, these data suggest the safety of HSV-DICP34.5 in PLWH might be tolerable. We have added the corresponding description in the revised manuscript.

| (6) *Discuss the potential of inflammatory HSV vaccines to be used in PLWH without clinical symptoms.*

Thank you for your mention. As discussed above, we found that HSV-DICP34.5 exhibited lower virulence and replication ability than its parental strain (Figure 1D, Figure S1), and we also found that HSV-DICP34.5 induced a lower level of inflammatory cytokines (including IL-6, IL-1 $\beta$ , and TNF- $\alpha$ ) in primary CD4<sup>+</sup> T cells from PLWH compared to HSV-GFP stimulation, likely due to its lower virulence and replication ability (Figure 1I-K). In addition, the CD4<sup>+</sup>/CD8<sup>+</sup> T cell ratio (Figure 5I) and body weight (Figure S9) after treatment were effectively

ameliorated in the SIV-infected macaques of the ART+HSV-DICP34.5-sPD1-SIVgag/SIVenv group. Our data also demonstrated that there was no significant effect on the cell composition of peripheral blood in the SIV-infected macaques of ART+HSV-sPD1-SIVgag/SIVenv group (Figure S10). Thus, these data suggest the safety of HSV-DICP34.5 in PLWH might be tolerable. We have added the corresponding description in the revised manuscript.

**Reviewer #2 (Recommendations For The Authors):**

*I think the authors have done due diligence to the experimental system, and collected evidence to show the feasibility of delaying virus rebound in macaques. However, I would encourage the authors to perform experiments that can back up the claim that delayed virus rebound is due to neutralization effects, or perhaps due to a reduction in viral reservoir. I believe insights into this process will add rigor, and push the relevance of the study to the next level.*

Thank you for your nice comment and valuable suggestion. We have now provided more data about this issue. We found significant suppression of total SIV DNA and integrated SIV DNA provirus in the ART+HSV-sPD1-SIVgag/SIVenv group. However, the copies of the SIV DNA provirus were significantly improved in the ART+HSV-empty group and ART+saline group (Figure 5F-G). We also discussed that incorporating the induction of broadly neutralizing antibodies into our future optimizing approaches may lead to better therapeutic outcomes in the revised Discussion section. We have added the corresponding description in the revised manuscript. Thank you.

Altogether, all of the above comments and suggestions are very helpful in improving our manuscript. We have taken these comments into account seriously and try our best to address these questions point-by-point. After making extensive revisions, we now submit this revised manuscript for your re-consideration. Thank you again for all of your comments and suggestions.

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